Interpreting Order Effects

Quantum Cognition for Human-AI Decision Teaming

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M2S1

Order Differences and QQ

Which statistic compares F across orders?

Which statistic compares A/B agreement?

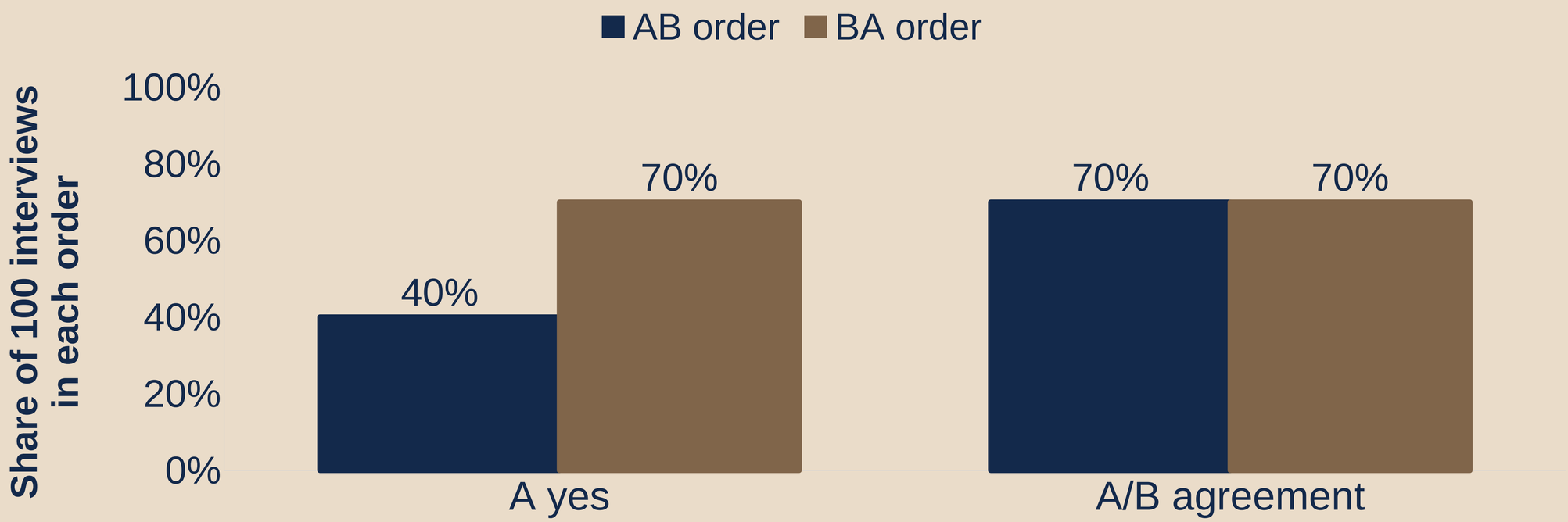
What records does each require?

An Imposed Answer-Pair Table

Answer pair	AB	BA
A yes, B yes	30	45
A yes, B no	10	25
A no, B yes	20	5
A no, B no	40	25

Both columns contain 100 illustrative interviews.

Agreement and the A-Yes Rate



Calculated from the imposed answer-pair table on the previous slide.

Why can agreement stay fixed while an answer rate changes?

A Classical Sequential Explanation

Inspect the transition rules in the notebook.

The rules vary with which question comes first.

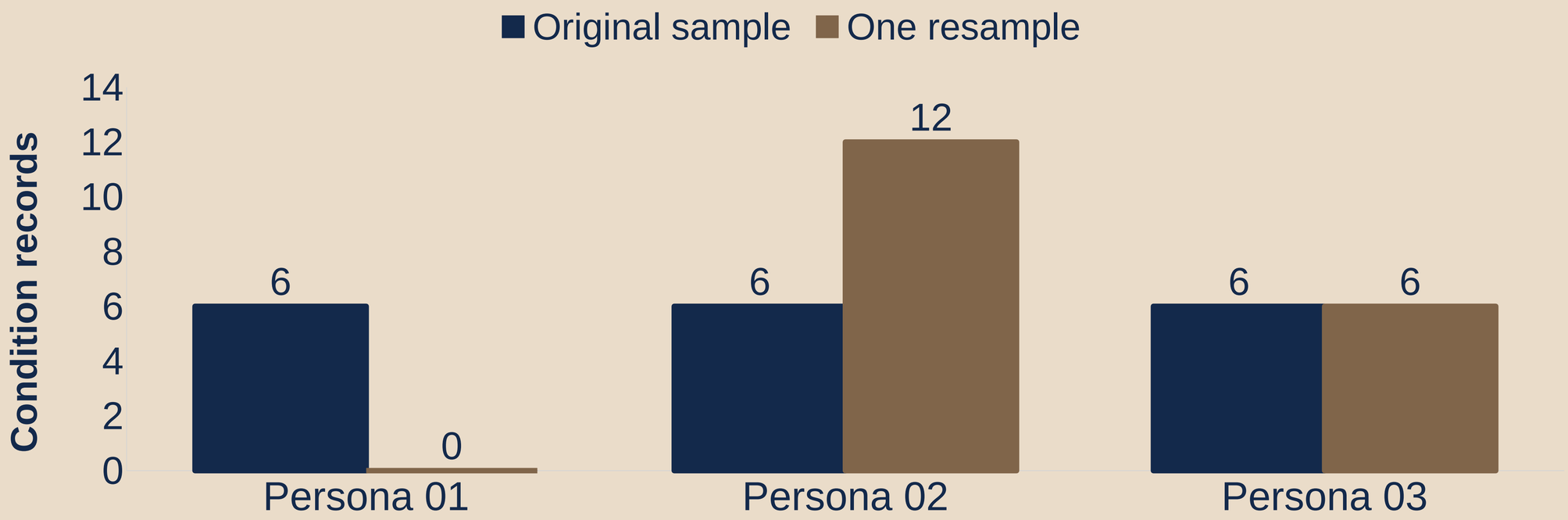
Which extra prediction would test this explanation?

Intervals Against a Practical Range

Case	Illustrative 95% interval	Decision relative to ±5 pp
A	−2 to +3 pp	Within the range
B	−12 to +14 pp	Unresolved
C	+1 to +4 pp	Within the range

Use a prespecified tolerance of ±5 percentage points.

Resampling Complete Personas



Toy example: draw personas 02, 02, 03 from three complete blocks.

Which six records travel together when a persona is drawn?

Design Revisions

Freeze A, B, F, coding, and collection conditions.

Choose a practical threshold and q tolerance.

Write a fair test of the strongest alternative.

The Interpretation Paragraph

Explain why order and QQ answer different questions.

Describe what an uncertain interval leaves unresolved.

Use that reasoning in Discussion 2.